The reconstruction map is a left inverse of the map given by core and tableau

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Editorial note. The counted statement is the author's labelled lemma environment at publisher-source.tex lines 982-985; its proof environment opens at line 986, immediately after \end{lemma}, so the pairing is by adjacency and the retained proof range 986-995 is the whole environment.

Editorial note. Two statements are retained as context because the counted proof cites them: lem:properties of core, item (2) (source lines 873-881), used at source line 992, and obs:restriction of matching to set and complement defines uniquely (source lines 692-695), also used at source line 992.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label lem:F_tilde(F)=id.

Editorial note. r002 note: r001 failed with an undefined reference to def:restriction of matching (addressed by adding that definition as context) and with a LaTeX error because the retained observation context uses the author's observation environment, which the prepared presentation adapter of this source re-declares on the existing wrapper counter; r002 adds that one adapter line as a second preamble span.

Editorial scope. One original result of the pinned publisher article, counted once: the reconstruction map is a left inverse of the map given by core and tableau (author label lem:F_tilde(F)=id).

Allowed standard prerequisites (editorial).

- The maps F and F-tilde. F sends a matching to the pair formed by its core and its two-row tableau, and F-tilde reconstructs a matching from a core and a tableau.. They are the two maps composed in the statement.
- Stable set and restriction of a matching. The restriction of a matching to an m-invariant set of vertices is again a matching, and a matching is determined by its restrictions to a set and its complement.. They let the proof reduce the comparison to the stable and the unstable part.
- Unstable part as a function of the tableau. The unstable chords of a matching are reconstructed from the two-row tableau through the permutation p(T).. It is the equation proved at the start of the argument.

Definition 0.1. Given a matching $m \in \mathcal{M}_N$, we say that a set $S \subseteq [N]$ is m-invariant if for every chord $(i,j) \in m$ we have $i \in S$ if and only if $j \in S$. Moreover, for a matching m and an m-invariant set S we define the restriction of m to S, denoted $\mathrm{res}_S(m)$, to be a matching on S which is obtained by removing all the vertices not in S from m.

Lemma 0.2. The reduction process has the following properties:

(1) If a chord intersects another chord then it is stable.

(2) A chord (i,j) is stable if and only if there exists a stable vertex in $\{i+1,\ldots,j-1\}$.

(3) Given a matching m and $i<j$, if the set $[i,j]$ is m-invariant (as defined in Definition 0.1) and the restricted matching $\mathrm{res}_{[i,j]}(m)$ is perfect and non-crossing, then $\ell \notin \mathrm{Stable}(m)$ for all $\ell \in [i,j]$.

Observation 0.3. Let $S \subseteq [N]$ be a set of vertices, let m_1 be a matching on S, and let m_2 be a matching on $[N] \setminus S$. Then there exists a unique matching $m \in \mathcal{M}_N$ such that $\mathrm{res}_S(m) = m_1$ and $\mathrm{res}_{[N]\setminus S}(m) = m_2$.

Lemma 0.4. Let $m \in \mathcal{M}_{N,f}$ be a matching, and denote $F(m) = (\mathrm{core}(m), T)$. Then $\tilde{F}(\mathrm{core}(m), T) = m$.

Proof. Denote $|\mathrm{Stable}(m)| = N-2k$, implying that $\mathrm{core}(m) \in \mathcal{M}_{N-2k,f}$ and $T \in \mathrm{SYT}(N-k,k)$. Moreover, denote $p = p(T) \in \mathcal{P}_{N,N-2k}$. First, we prove that

$$\mathrm{res}_{[N]\setminus \mathrm{Stable}(m)}(m) = m_{\mathsf{unstable}}(p). \tag{0.1}$$

We note that the set $[N] \setminus \mathrm{Stable}(m)$ is m-invariant, so the left-hand side of Equation (0.1) is well-defined. Notice that $|[N] \setminus \mathrm{Stable}(m)| = |[N] \setminus \mathrm{Stable}(p)| = 2k$ and that $\mathrm{res}_{[N]\setminus \mathrm{Stable}(m)}(m)$ is a perfect matching. Therefore, in order to prove that Equation (0.1) holds, it suffices to prove that any unstable chord of m belongs to $m_{\mathsf{unstable}}(p)$ too. Let $(i,j) \in m$ be an unstable chord. Thus, we may deduce that $i \in \mathrm{UP}(p)$ and $j \in \mathrm{DOWN}(p)$. Let $i' \in \mathrm{UP}(p)$ such that $i < i' < j$. By Lemma 0.2 (part 2), i' is an endpoint of an unstable chord $(i',j') \in m$ with $i' < j'$. By Lemma 0.2 (part 1), we obtain that the chord (i,j) does not intersect (i',j'), implying that $i < i' < j' < j$. On the other hand, every $j' \in \mathrm{DOWN}(p)$ with $i < j' < j$ is an endpoint of an unstable chord $(i',j') \in m$ with $i < i' < j' < j$. Therefore, we obtain a bijection from $\mathrm{UP}(p) \cap [i,j]$ to $\mathrm{DOWN}(p) \cap [i,j]$ (with i matched with j), so $\mathrm{height}_p(i) = \mathrm{height}_p(j)$. Moreover, this bijection has the property that if $i' \in \mathrm{UP}(p) \cap [i,j]$ is matched with $j' \in \mathrm{DOWN}(p) \cap [i,j]$ then $i' < j'$ (i.e. the ascending steps of the path appear before the associated descending steps), so $\mathrm{height}_p(j) < \mathrm{height}_p(i')$ for all $i' \in \mathrm{UP}(p) \cap [i+1, j-1]$. We may conclude that $(i,j) \in m_{\mathsf{unstable}}(p)$ for every unstable chord (i,j) of m and prove that Equation (0.1) holds.

Next, we denote $m' = \tilde{F}(\mathrm{core}(m), T)$ and prove that $m = m'$. From Equation (0.1) we deduce that the supports of the matchings $\mathrm{res}_{[N]\setminus \mathrm{Stable}(m)}(m)$ and $m_{\mathsf{unstable}}(p)$ are identical, and therefore $\mathrm{Stable}(m) = \mathrm{Stable}(p)$. Thus, the set $[N] \setminus \mathrm{Stable}(m)$ is both m-invariant and m'-invariant, and restricting each of these matchings to $[N] \setminus \mathrm{Stable}(m)$ results in $m_{\mathsf{unstable}}(p)$. In addition, it can be easily verified from the descriptions of F and $\tilde{F}$ that $\mathrm{res}_{\mathrm{Stable}(m)}(m) = \mathrm{res}_{\mathrm{Stable}(m)}(m')$ is the matching obtained by relabeling the vertices of $\mathrm{core}(m)$ with the elements of $\mathrm{Stable}(m)$ in increasing order. By Observation 0.3, we may deduce that $m = m'$. □